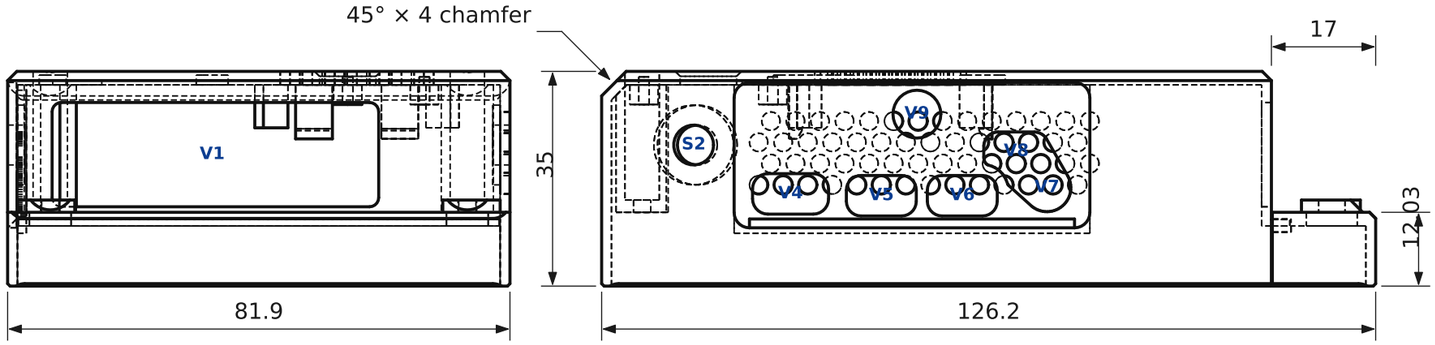
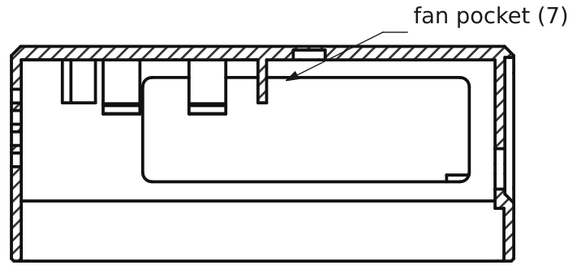
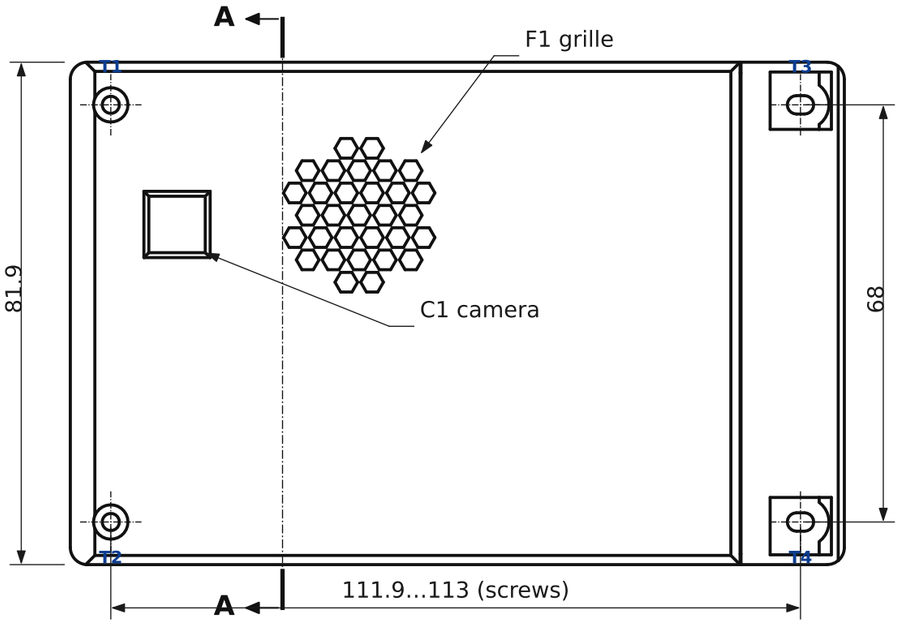


SIDE VIEW (I/O face, +x) FRONT VIEW (HDMI side, -y)



SECTION A-A (x = -28.5)



REAR VIEW (plan, +z)

OPENINGS · origin = display PCB centre, z = 0 at the glass face

ID	Face	Feature	Centre 1	Centre 2	Size
V1	+x	USB + Ethernet (1 window)	y -33.7...19.55	z 21.43	53.25 × 17 R1.5
B1	-y	Recessed bay 1.45 (wall 1.6)	x -41.5...16.5	z 9.5...33.2	R2.5
V4	-y	USB-C (power)	x -32.3	z 15.03	12.4 × 6.6 R2.5
V5	-y	micro-HDMI 0	x -17.5	z 14.73	11.4 × 6.6 R2.4
V6	-y	micro-HDMI 1	x -4.3	z 14.73	11.4 × 6.6 R2.4
V7+8	-y	Pi audio + HAT micro-USB	x 9.5/4.5	z 16.03/21.93	1 opening · 1.8 wall to V6
V9	-y	4G HAT audio	x -11.7	z 28.03	Ø7.8
S1	+y	4G antenna SMA (MAIN)	x -47.5	z 23	Ø6.5 + Ø13 boss
S2	-y	4G antenna SMA (AUX)	x -48.1	z 23	Ø6.5 + Ø13 boss
C1	+z	Camera v1.3 window	x -45.7	y 14.5	9.2 × 9.2 + 0.8 chamfer
F1	+z	Grille + 4 fan clips	x -16	y 16	hex 3.2 · 4 hooks · 3 guides
T1-2	-x	Screw to display standoff	x -56.5	y ±34	Ø2.8 · c-bore Ø5.6 to z 14.03
T3-4	+x	P6-03 frame screw to standoff	x 55.3...56.6	y ±34	Ø3 slot · 2 mm pad

NOTES

Outline 126.2 × 81.9 · R2.8 · overall thickness 35.0.
Display pocket 123 × 78.7 · R1 (clearance 1 / 0.75).
I/O lip = separate P6-03 frame; lap joint: 3 × 1.05 tongue.
Side wall 1.6 · rear 2.2 · pogo pockets 1.6 · capacitor 0.8.
Glass flush with the rim, standoff tops at z 12.03.
Fixing: 4 × M2.5 × 6 into the display corner standoffs
(I/O side through slots, ±0.65).
Camera: 4 bosses Ø4.6 × 3.2 · M2 self-tapping.
Fan 30×30×7 flush with the rear face, clips only:
4 hooks (1.3 arm, 0.8 barb) + 3 corner guides.
Exhaust: Ø3 holes (56 on +y, 65 on -x) + I/O window.
Dimensions marked (*) in params_v6.py: confirm with P6-00.
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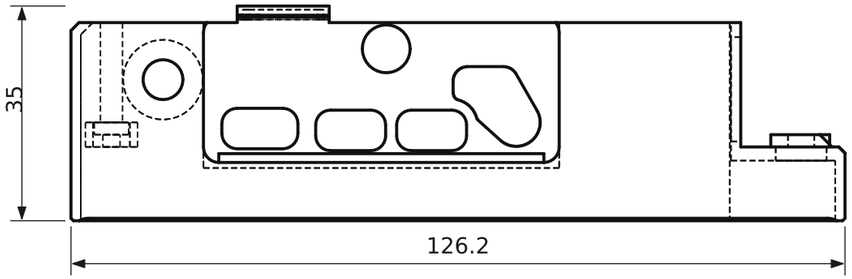
LITTLE RASCAL · PANEL5 5″ DSI · Pi 4B · 4G — V6 "SANDWICH"

No supports · P6-03 prints standing up

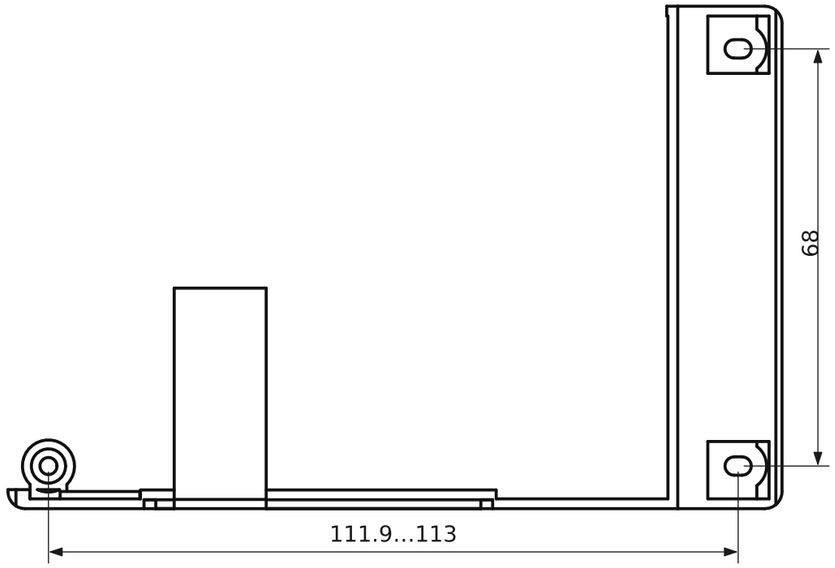
TITLE	PART NO.
V6 case + I/O frame	P6-01/03 V6.9

MATERIAL	GENERAL TOL.	DATE	SIZE
PETG	ISO 2768-m	2026-10-27	A3

SCALE	UNITS	PROJECTION	SHEET
1:1	mm		1/2



FRONT VIEW (-y)



PLAN (+z)

CHECKS AND HEIGHTS

USING THE TEMPLATE (before printing P6-01):
1. Assemble the full sandwich (Pi + HAT).
2. Screw the template on with 3 × M2.5 × 6
   (2 on the lip, 1 at the -x/-y corner).
   If they do not line up: note the offset.
3. Every port should look centred in its
   window, with ≥ 0.5 mm clearance.
4. The lower edge of the -y wall should sit
   flush with the glass.
5. The pogo pins must fit under their
   pockets (1.6 mm; the spring tip retracts).
6. If something does not fit, note the offset,
   adjust PI_X0 / PI_Y0 / BRASS / Z_POGO in
   params_v6.py and regenerate the case.

SANDWICH HEIGHTS (z from the glass):
glass 0 · display PCB 6.4–8
standoffs 8–12.03 · Pi PCB 12.03–13.43
USB 29.43 · HAT 23.43–25.03 · HAT USB-A 32.03
pogo fixed part 34.03 · inner rear face 32.8

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BILL OF MATERIALS

#	Component	Qty	Note
1	Waveshare 5inch DSI LCD (C) 800×480 display	1	own M2.5 standoffs
2	Raspberry Pi 4B	1	on the 4 display standoffs
3	Waveshare SIM7600G-H 4G HAT (B)	1	M2.5 × 10 brass standoffs
4	Raspberry Pi Camera v1.3 + 15-pin FFC	1	150 mm recommended (min. ~90)
5	Fan 30×30×7 mm 5 V	1	snaps into the clips
6	SMA female bulkhead + u.FL pigtail	2	MAIN and AUX; nut outside
7	4G antenna, SMA male	2	
8	M2.5 × 6 screw (ISO 7380 or DIN 912)	4	case/frame → corner standoffs
9	M2 × 5 self-tapping screw (nylon ideal)	4	camera
10	Micro-USB 90° ↔ USB-A 90° cable, 10–15 cm	1	HAT V8 → Pi USB 2.0 (V1)
11	Adhesive rubber feet Ø8, ≥ 3 mm	4	optional
12	Printed: P6-01 case + P6-03 frame (P6-00/02 tests)	1+1	PETG

LITTLE RASCAL · PANEL5 5″ DSI · Pi 4B · 4G — V6 "SANDWICH"

Print on its -y face · ~1 h · 15 cm³

TITLE	PART NO.
Fit-check template	P6-00 V6.9

MATERIAL	GENERAL TOL.	DATE	SIZE
PLA / PETG	ISO 2768-m	2026-10-27	A3

SCALE	UNITS	PROJECTION	SHEET
1:1	mm		2/2